Answer Key - Engineering - Graduate Level Assessment 11 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** A
   Options: A) VarZ = VarX + VarY - 2 E[{X - E(X)}{Y - E(Y)}]; B) VarZ = |VarX - VarY|; C) VarZ =VarX +VarY; D) VarZ = VarX + VarY + Cov(X,Y); E) VarZ = (VarX + VarY) / 2
   Note: Correct option: A - VarZ = VarX + VarY - 2 E[{X - E(X)}{Y - E(Y)}]

2. **Answer:** A
   Options: A) extend the motor's lifespan.; B) increase the motor's efficiency.; C) reduce the motor's noise.; D) maintain a constant speed.; E) improve the motor's torque.
   Note: Correct option: A - extend the motor's lifespan.

3. **Answer:** A
   Options: A) 2121.32 A.; B) 6000 A.; C) 2000 A.; D) 7000 A.; E) 5000 A.
   Note: Correct option: A - 2121.32 A.

4. **Answer:** A
   Options: A) 5/36; B) 3/12; C) 6/36; D) 1/6; E) 17/36
   Note: Correct option: A - 5/36

5. **Answer:** A
   Options: A) 100 ohms; B) 0 ohms; C) - 50 j ohms; D) -100 j ohms; E) + 25 j ohms
   Note: Correct option: A - 100 ohms

True / False

6. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: '25 ohms'.

7. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

8. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: '15.38 psf'.

9. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: 'E(1) = 2 J, E(2) = 4 J, E(3) = 6 J, E(4) = 8 J, E(5) = 10 J, E(6) = 12 J'.

10. **Answer:** True
    Options: A) True; B) False
    Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 2.585 bits. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
    Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** E = - (1/2)bB_0 $e^{b}tra\_$ extphi. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
    Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key